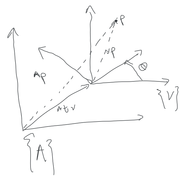
Transformation

Rotation + Translation.



$${}^{V}P = {}^{V}R_B \, {}^{B}P$$

$${}^{A}P = {}^{A}t_V$$
$$= {}^{A}t_V + {}^{V}R_B \, {}^{B}P$$

$${}^{A}P = {}^{A}t_B + {}^{A}R_B \, {}^{B}P$$

since $\{V\} \| \{A\}$

So $V = A$

$$\begin{bmatrix} A_x \\ A_y \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} {}^{B}X \\ {}^{B}Y \end{bmatrix} \begin{bmatrix} {}^{A}X_B \\ {}^{A}Y_B \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta & -\sin\theta & {}^{A}X_B \\ \sin\theta & \cos\theta & {}^{A}Y_B \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} B_x \\ B_y \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} {}^{A}R_B & {}^{A}t_B \\ 0,0 & 1 \end{bmatrix} {}^{B}P$$

$$\text{Rotation}^{-1} = \text{Rotation}^{t}$$

$$\text{Transformation}^{-1} \neq \text{Transformation}^{t}$$